

Assignment 10

Dockerize & Deploy Express.js App
using Docker Compose with Nginx

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My GitHub repo: https://github.com/MahmudurRahman36/Mahmud_Assignment10

DockerHub: https://hub.docker.com/r/mrkolince/mahmud_assignment10

Source repo: <https://github.com/roy35-909/Module-3-deployment>

Step 1 - Clone repository & project structure

```
C:\WINDOWS\system32\conhost.exe
PS> git clone https://github.com/roy35-909/Module-3-deployment
PS> cd Module-3-deployment
PS> ls

Mode                LastWriteTime         Name
----                -
d-r--- 6/23/2026 11:38:33 PM .github
d-r--- 6/23/2026 11:38:33 PM src
d-r--- 6/23/2026 11:38:33 PM test
-a---- 6/23/2026 11:38:33 PM .gitignore
-a---- 6/24/2026 12:15:51 AM docker-compose.yml
-a---- 6/23/2026 11:40:01 PM Dockerfile
-a---- 6/23/2026 11:38:33 PM eslint.config.js
-a---- 6/23/2026 11:38:33 PM index.html
-a---- 6/23/2026 11:38:33 PM Jenkinsfile
-a---- 6/23/2026 11:38:33 PM package-lock.json
-a---- 6/23/2026 11:38:33 PM package.json
-a---- 6/23/2026 11:38:33 PM postcss.config.js
-a---- 6/23/2026 11:38:33 PM README.md
-a---- 6/23/2026 11:38:33 PM tsconfig.app.json
-a---- 6/23/2026 11:38:33 PM tsconfig.json
-a---- 6/23/2026 11:38:33 PM tsconfig.node.json
-a---- 6/23/2026 11:38:33 PM vite.config.ts

PS D:\1_Office_Document\4. Training\DevOps\Ostad\Assignment10\Mahmud_Assignment10\forupload\Module-3-deploy
```

Step 2 - Dockerfile (containerizes the Express.js app)

```
C:\WINDOWS\system32\conhost.exe
STEP 2 : The Dockerfile (containerizes the Express.js app)
-----

PS> cat Dockerfile

FROM node:18-alpine

WORKDIR /app

COPY package*.json ./

RUN npm install --production

COPY . .

EXPOSE 5000

ENV PORT=5000

CMD ["npm", "start"]
PS D:\1_Office_Document\4. Training\DevOps\Ostad\Assignment10\Mahmud_Assignment10\forupload\Module-3-deploy
```

Step 3 - docker-compose.yml (Nginx + Express, app after Nginx)

```
C:\WINDOWS\system32\conhost.exe
STEP 3 : The docker-compose.yml (Nginx + Express, app starts after Nginx)
-----
PS> cat docker-compose.yml

services:
  nginx:
    image: nginx:latest
    container_name: nginx-server
    restart: always

  app:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: express-app
    ports:
      - "8080:5000"
    environment:
      - PORT=5000
    depends_on:
      - nginx
    restart: always
PS D:\1_Office_Document\4. Training\DevOps\Ostad\Assignment10\Mahmud_Assignment10\forupload\Module-3-deploy
```

Step 4 - docker build -t mrkolince/mahmud_assignment10:latest .

C:\WINDOWS\system32\conhost.exe

STEP 4 : Build the Docker image

PS> docker build -t mrkolince/mahmud_assignment10:latest .

```
[+] Building 2.9s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 186B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9
=> [internal] load build context
=> => transferring context: 3.33kB
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY package*.json ./
=> CACHED [4/5] RUN npm install --production
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:0f03c0f00b704de27517e8c26ed95d247fae2d6c64393cb5c9003cfd9ddf7f19
=> => exporting config sha256:1ede48c4c8e86116ed2eb4899bf6353d840c4add73494120960716730e99051c
=> => exporting attestation manifest sha256:63095c2abb702086c60946c665d5fb6f36b16638d3148e77401469b63acf7c
=> => exporting manifest list sha256:62ee5a9b327951078e61b5e9ae1f55316d02c93d2e5552ce0aa57c2c1eb51730
=> => naming to docker.io/mrkolince/mahmud_assignment10:latest
=> => unpacking to docker.io/mrkolince/mahmud_assignment10:latest
```

PS> docker images mrkolince/mahmud_assignment10

IMAGE	ID	SIZE	IMAGE	CONTENT	SIZE	EXTRA
-------	----	------	-------	---------	------	-------

Step 5 - docker push mrkolince/mahmud_assignment10:latest

C:\WINDOWS\system32\conhost.exe

STEP 5 : Push the image to DockerHub

PS> docker login (already logged in as mrkolince)

PS> docker push mrkolince/mahmud_assignment10:latest

The push refers to repository [docker.io/mrkolince/mahmud_assignment10]

dd71dde834b5: Layer already exists

7a15289fc279: Layer already exists

38a75d7ef1f7: Layer already exists

1e5a4c89cee5: Layer already exists

5c38fd8216b5: Layer already exists

f18232174bc9: Layer already exists

1b47c26764a0: Pushed

edcf5ae64164: Pushed

25ff2da83641: Layer already exists

latest: digest: sha256:62ee5a9b327951078e61b5e9ae1f55316d02c93d2e5552ce0aa57c2c1eb51730 size: 856

PS D:\1_Office_Document\4. Training\DevOps\Ostad\Assignment10\Mahmud_Assignment10\forupload\Module-3-deploy

Step 6 - docker compose up -d

C:\WINDOWS\system32\conhost.exe

[+] up 10/10

✓ Image nginx:latest Pulled

#1 [internal] load local bake definitions

#1 reading from stdin 697B done

#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile

#2 transferring dockerfile: 186B done

#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/node:18-alpine

#3 DONE 1.0s

#4 [internal] load .dockerignore

#4 transferring context: 2B done

#4 DONE 0.0s

#5 [internal] load build context

#5 transferring context: 3.00kB 0.0s done

#5 DONE 0.0s

#6 [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9b

#6 resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9b4b

#6 DONE 0.0s

#7 [4/5] RUN npm install --production

#7 CACHED

#8 [2/5] WORKDIR /app

#8 CACHED

Step 7 - Verify running containers (docker ps)

C:\WINDOWS\system32\conhost.exe

STEP 7 : Verify the running containers

PS> docker compose ps

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
express-app	module-3-deployment-app	"docker-entrypoint.s..."	app	30 seconds ago	Up 29 seconds
nginx-server	nginx:latest	"/docker-entrypoint..."	nginx	30 seconds ago	Up 30 seconds

PS> docker ps

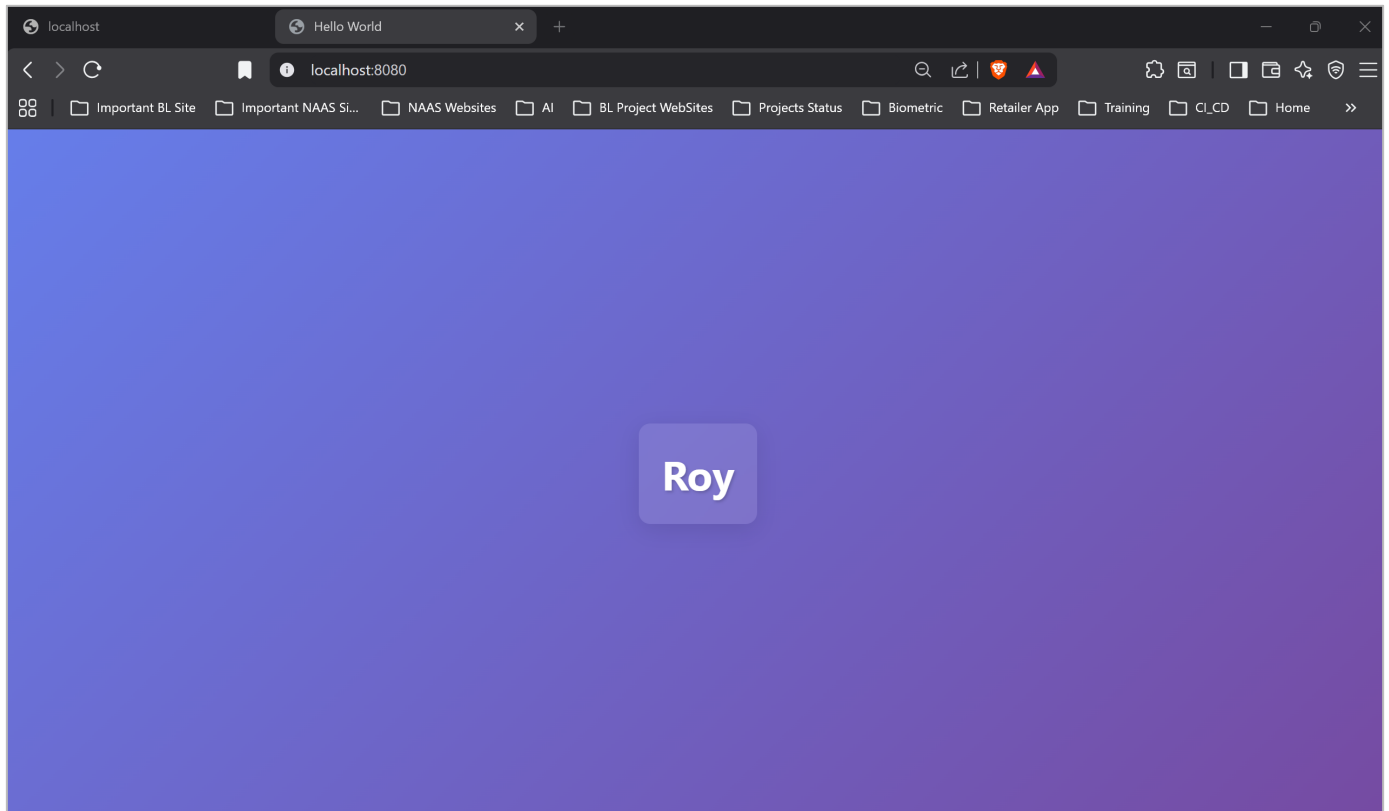
NAMES	IMAGE	STATUS	PORTS
express-app	module-3-deployment-app	Up 30 seconds	0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp
nginx-server	nginx:latest	Up 30 seconds	80/tcp

Express app -> http://localhost:8080 (host 8080 -> container 5000)

Nginx -> nginx:latest running, port 80/tcp (internal)

PS D:\1_Office_Document\4. Training\DevOps\Ostad\Assignment10\Mahmud_Assignment10\forupload\Module-3-deploy

Step 8 - App working in browser at http://localhost:8080



Step 9 - Image published on DockerHub (latest tag)

mrkolince
Your personal account

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Upgrade plan

Repositories / mahmud_assignment10 / General

mrkolince/mahmud_assignment10

Last pushed 33 minutes ago · Repository size: 44.7 MB · 0 stars · 36 downloads

Add a description

Add a category

Docker commands

To push a new tag to this repository:

```
docker push mrkolince/mahmud_assignment10:tag name
```

Public view

General

Tags

Image Management

Collaborators

Webhooks

Settings

Tags

DOCKER SCOUT INACTIVE

Activate

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	33 minutes

See all

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Step 10 - Compose stack running in Docker Desktop

 docker.desktop

PERSONAL

Ctrl+K



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 Gordon

 Containers

 Images

 Volumes

 Kubernetes

 Builds

 Models

 MCP Toolkit BETA

 Docker Hub

 Logs

 Extensions

Containers [Give feedback](#)

Container CPU usage ⓘ
0.00% / 1800% (18 CPUs available)

Container memory usage ⓘ
48.89MB / 7.31GB [Show charts](#)

 ☐ Only show running containers

☐		Name	Container ID	Image	Port(s)	CPU (%)	Memc	Actions
☐	▼	module-3-deplo	-	-	-	0%	48.89	   
☐		nginx-server	9495b5ca5984	nginx:lates		0%	15.99	   
☐		express-app	a93ea9dc4aa7	module-3-c	8080:5000 	0%	32.9M	   

Upgrade plan

Preparing to install update. 

 Engine running

RAM 1.20 GB CPU 0.06% Disk: 1.93 GB used (limit 1006.85 GB)

 Update version